

FocusParse: Evidence Packets for High-Resolution Document Question Answering

Anonymous submission

Abstract

High-resolution technical and financial PDFs expose a persistent failure mode for multimodal document question answering: whole pages preserve context but compress away small labels, footnotes, and chart/table details, while tight crops recover readability but detach evidence from the local context needed to interpret it. We study this failure mode using parser-bench, a citation-required visual QA benchmark over finance reports and technical datasheets where answers are evaluated with both correctness and page/box grounding. We introduce FocusParse, a structured harness that routes to likely pages, localizes candidate regions, inspects high-resolution crops, expands selected evidence with role-labeled neighboring context, and answers only from compact evidence packets. In a matched seven-method run on the pinned 148-row paper slice, FocusParse +4 reaches 61.5% overall accuracy, 66.3% datasheet accuracy, and 51.1% finance accuracy, compared with 43.9% for a base VLM and 6.1–18.2% for generic tool-agent comparators. The same run shows high page recall (0.914), strong region overlap (0.857 mean BBox IoU), and low lazy answering (3.4%). The resulting thesis is that performance on dense finance and datasheet documents depends not merely on bigger context windows or more tools, but on constructing the right evidence unit before reasoning: a readable region plus the captions, legends, headers, footnotes, axes, and cross-page references that make the region meaningful.

1 Introduction

Document AI systems increasingly need to answer questions over high-resolution PDFs rather than simply transcribe them. In finance reports and technical datasheets, the answer often depends on small plotted labels, multi-panel figures, footnotes, min/typ/max tables, timing diagrams, and symbol dictionaries spread across multiple pages. This creates a resolution-context tradeoff. A full-page VLM call can preserve surrounding layout but may make fine details unreadable. A tight crop can make the relevant text or visual element readable but may lose the caption, legend, table header, or footnote that determines what the crop means.

FocusParse is built around the claim that the useful intermediate object for this setting is an *evidence packet*. An evidence packet binds a localized region to its supporting context: local crop, text/OCR snippets, role-labeled neighbor crops, provenance, confidence, and optional extracted structure. The method is deliberately narrower than a general-purpose parser. It is a harness for query-conditioned evidence construction in dense documents where the answer must be tied to supporting pages and boxes.

This paper makes four contributions. First, it frames parser-bench as a region-grounded QA setting for concentrated finance and datasheet documents. Second, it describes FocusParse, an evidence-localization-first harness with explicit inspect and expand stages. Third, it compares base VLMs, generic ReAct loops, generic tool agents, and FocusParse under matched tools and scorers. Fourth, it analyzes answer accuracy, cost per correct, page recall, BBox IoU, lazy-answer rate, and qualitative packet traces.

2 Related Work

Document parsing benchmarks show that document pages contain structured visual regions and that parsing quality should be evaluated beyond plain OCR [9, 15, 14, 8, 16, 12]. MPDocBench-Parse is useful because it moves parsing evaluation toward practical multi-page documents, semantic continuity, hierarchy recovery, and visual content preservation. External ParseBench is especially relevant because it evaluates agent-facing parsing along tables, charts, content faithfulness, semantic formatting, and visual grounding. FocusParse

Table 1: Canonical parser-bench paper slice used for the final matched run.

Slice	Rows	Unique IDs	PDFs	Datasheet	Finance
Canonical paper slice	148	147	44	101	47

asks a different but adjacent question: not whether a page or multi-page document can be fully parsed, but whether a QA system can select the answer-supporting evidence for a query, attach the local context needed to interpret it, and cite the supporting page/box. Throughout this draft, **ParseBench** refers to the RunLlama parsing benchmark, while **parser-bench** refers to the Gabriel benchmark used for FocusParse evaluation.

Document QA and financial QA benchmarks establish the need for multimodal and domain-specific reasoning. DocVQA, InfographicVQA, and MMLongBench-Doc study questions over document images and long multimodal contexts [6, 5, 3]. FinQA, TAT-QA, FinanceBench, and FinRAGBench-V emphasize numerical reasoning, open-book financial QA, and visual citations [1, 18, 2, 13]. Parser-bench differs by making high-resolution region localization, multi-piece context collection, and page/crop citation part of the QA contract.

Chart and table QA benchmarks such as ChartQA and PlotQA show that charts require both visual perception and reasoning [4, 7]. FocusParse places this challenge inside long PDFs where chart interpretation often depends on neighboring captions, axes, legends, footnotes, or table references. The chart-reading problem is therefore inseparable from evidence collection: the model must read the mark and recover the nearby semantics that make the mark meaningful.

The closest methodological neighbors are AgenticOCR and DocLens. AgenticOCR argues that page-level chunking introduces extraneous context and dilutes salient evidence, reframing OCR as query-conditioned on-demand extraction [10]. DocLens identifies evidence localization as a central failure mode in long visual document understanding and navigates from documents to pages and elements before answer adjudication [17]. FocusParse shares the active evidence-acquisition motivation, but studies a controlled stage machine with explicit typed evidence packets and role-labeled context expansion. The contrast is the object handed to the reasoner: a compact packet containing a readable crop plus captions, legends, headers, footnotes, axes, or neighboring regions, rather than a loose transcript of tool observations.

Generic ReAct-style agents provide a natural baseline for tool use [11]. The key comparison is not tools versus no tools. It is unconstrained tool use versus structured evidence construction. Generic loops may inspect documents, but in dense PDFs they can also terminate lazily, repeat irrelevant calls, or answer from noisy transcripts that contain weak citations.

3 Parser-Bench Dataset

Parser-bench is a citation-required visual QA benchmark over full-resolution finance reports and technical datasheets. Each example contains a question, a gold answer, answer-type metadata, source document metadata, rendered page images, supporting pages, supporting bounding boxes, optional alternate boxes, question-family labels, difficulty axes, and reasoning-chain metadata. The current paper slice is the canonical post-filter validation materialization: 148 rows, 147 unique IDs, 44 source PDFs, 101 datasheet examples, and 47 finance examples.

The slice is small but deliberately concentrated. It contains 70 multi-region examples, 34 multi-page examples, 112 visually grounded examples, and 49 rows that are both visual and multi-region. The most frequent question families are axis-value interpolation, near-miss distractors, cross-page continuation, min/typ/max disambiguation, and chart/table cross-reference. These are exactly the families where inspection and context expansion matter: visual labels must be read at high resolution, then connected to surrounding document semantics.

The current pinned materialization records a fixed Hugging Face revision, dataset fingerprint, and benchmark JSONL SHA-256 in the artifact manifest. The full identifiers are reported with the released evaluation package rather than treated as moving URLs.

Table 2: Final matched seven-method result on the pinned 148-row parser-bench slice.

Method	Overall	Datasheet	Finance	\$/correct	Page recall	BBox IoU
Base VLM	43.9	47.5	36.2	0.0102	0.848	0.000
ReAct +2	18.2	22.8	8.5	0.1341	0.819	0.336
ReAct +4	16.2	20.8	6.4	0.1946	0.727	0.309
Agent baseline +2	8.1	10.9	2.1	0.1324	0.000	0.000
Agent baseline +4	6.1	7.9	2.1	0.1811	0.000	0.000
FocusParse +2	60.1	64.4	51.1	0.0282	0.929	0.897
FocusParse +4	61.5	66.3	51.1	0.0248	0.914	0.857

4 Method: FocusParse

FocusParse is an eight-stage workflow:

PLAN → ROUTE_PAGES → LOCALIZE → RERANK → INSPECT → EXPAND_CONTEXT → ANSWER → VERIFY.

The stages communicate through typed events rather than arbitrary scratchpad state. Planning predicts question family and evidence types. Routing chooses candidate pages. Localization and reranking propose visual/textual regions. Inspection turns selected regions into readable evidence by cropping, extracting text/OCR, and optionally using region-specific tools. Expansion attaches bounded context such as captions, legends, headers, axes, footnotes, and nearby table text. Answering receives evidence packets rather than raw pages. Verification checks whether the answer is supported, whether citations are grounded, and whether a bounded repair action is needed.

The central contract is **EvidencePacket**. A packet contains page, normalized box, region type, provenance, crop references, text-layer and OCR snippets, linked neighbor crop references, neighbor-role labels, optional chart/table extraction, confidence, and selection metadata. This design creates an explicit bottleneck: the reasoner can only use compact evidence assembled by the upstream stages. That bottleneck is the mechanism the paper tests.

5 Experiments

The final matched comparison has seven rows: a base VLM, ReAct +2 tools, ReAct +4 tools, generic agent +2 tools, generic agent +4 tools, FocusParse +2 tools, and FocusParse +4 tools. The +2 tool set exposes `inspect_region` and `get_text_layer`; the +4 tool set adds `expand_context` and `run_python`. Metrics include answer accuracy with 95% bootstrap confidence intervals, cost per correct answer, mean latency, page recall, mean BBox IoU, lazy-answer rate, and qualitative packet traces.

6 Results

FocusParse +4 improves over the base VLM by 17.6 overall points, 18.8 datasheet points, and 14.9 finance points. It improves over generic ReAct and generic agent baselines by 43–55 overall points while remaining much cheaper per correct than those tool-agent comparators. Base VLM remains cheapest per correct, so the claim is an accuracy and localization gain, not a lowest-cost claim.

Diagnostics show that FocusParse +4 builds 1184 evidence packets, calls `expand_context` on every example, attaches 9.72 neighbors on average, and cites packets with 100% text coverage and 84.4% linked-context coverage. Generic ReAct calls tools but does not construct evidence packets, while the generic agent baselines mostly terminate without useful tool use.

6.1 Mechanism Ablations

Table 3 summarizes the current n=148 mechanism ablations. The strongest positive signal is query-conditioned reranking: restoring rerank improves accuracy from 56.1% to 61.5% and substantially improves grounding.

Table 3: Mechanism ablations against the final FocusParse +4 row.

Condition pair	Baseline	Full +4	Δ	Rec.	Reg.
No rerank $\rightarrow$ rerank	56.1	61.5	+5.4	19	11
No <code>expand_context</code> $\rightarrow$ expand	59.5	61.5	+2.0	14	11
Repair off $\rightarrow$ repair on	62.8	61.5	-1.4	9	11
Answer-shape repair off $\rightarrow$ on	64.2	61.5	-2.7	9	13

Table 4: Primary failure buckets for the final FocusParse +4 run.

Failure bucket	Count	Datasheet	Finance	Error share
Verifier unsupported	26	16	10	45.6
Partial localization	12	6	6	21.1
Localization miss	8	5	3	14.0
Reasoning/extraction	7	4	3	12.3
Lazy/no bbox	4	3	1	7.0

Restoring `expand_context` gives a smaller +2.0 point gain. In contrast, restoring verifier-directed repair moves 62.8% to 61.5% in this matched run, and restoring answer-shape repair moves 64.2% to 61.5%, so the repair mechanisms are best treated as mixed or negative controls rather than sources of the paper gain.

6.2 Failure Taxonomy

The final FocusParse +4 row is correct on 91/148 examples and incorrect on 57/148 examples. Table 4 summarizes the primary error buckets computed from the final `per_example.jsonl`.

The dominant remaining error is not generic tool laziness: only 4/57 errors are lazy/no-bbox failures. The largest bucket is verifier-unsupported answers after evidence collection, followed by incomplete or missed localization. This suggests that the next lift is likely better verification, evidence adjudication, and multi-region localization rather than simply adding tools.

7 Qualitative Analysis

The qualitative examples make the main metric story inspectable. In the Latvia finance example, the answer turns on linking chart labels from pages 108 and 109 to the LV $\rightarrow$ Latvia country-code table on page 8. FocusParse +4 has page recall 1.0, BBox IoU 0.99999, and three citations. ReAct +2 recovers the answer but with weak region grounding, while generic agents either choose Estonia or emit invalid raw transcripts. In the JESD204B datasheet example, baselines can sometimes state the answer *6*, but FocusParse is the row that preserves complete page recall and near-perfect region overlap across the timing and block-diagram evidence.

8 Limitations

The current dataset slice is small and concentrated: 148 rows with 47 finance examples. The paper therefore reports confidence intervals and avoids broad claims about all document QA. A stronger 66.9% FocusParse checkpoint is documented in research notes, but the raw run directory is not present in the current checkout; it is not used as a headline claim here. The qualitative panels and failure-taxonomy table are draft analysis artifacts; final styling depends on the selected venue template and page budget. Zotero export is currently blocked because Zotero Desktop/local API is unavailable on this profile. Finally, FocusParse is a research harness, not a general document parser or production service.

Finance evidence binding: charts + abbreviation table

Prediction: Latvia | pages 8, 108, 109 | page recall 1.0 | BBox IoU 0.99999

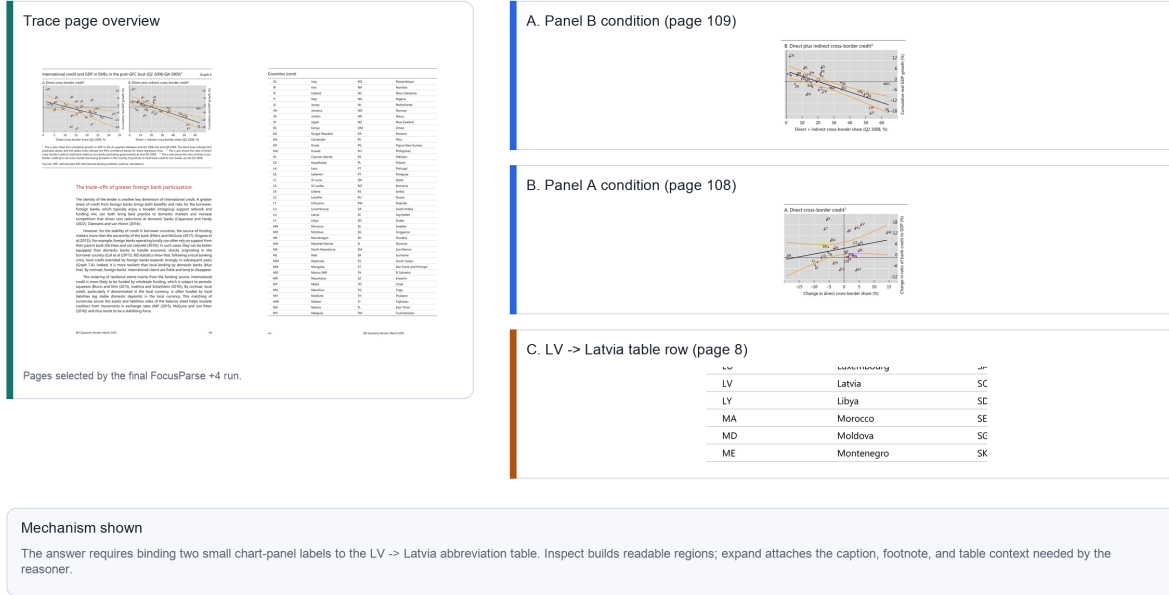


Figure 1: Finance evidence binding. The answer requires joining two chart-panel conditions to a country-code table. Same-revision baselines expose the failure mode: Base VLM returns unanswerable, generic agents often choose Estonia or emit raw tool transcripts, while FocusParse +4 answers *Latvia* with all three cited pages and near-perfect region overlap.

9 Conclusion

FocusParse studies a concrete mechanism for high-resolution document QA: construct the evidence before asking the model to answer. On the pinned 148-row parser-bench paper slice, FocusParse +4 reaches 61.5% accuracy with strong page and region grounding, outperforming both a no-tool base VLM and generic tool-agent comparators. The more important mechanism signal is that structured inspect and expand stages produce compact packets that preserve both readability and interpretive context. This supports the paper’s central claim: for dense finance and datasheet documents, better harnesses are not just models with larger context windows or more tools. They are systems that build the right evidence object before reasoning.

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Datasheet evidence compaction: timing count + path corroboration

Prediction: 6 | pages 16, 73 | page recall 1.0 | BBox IoU 0.99999

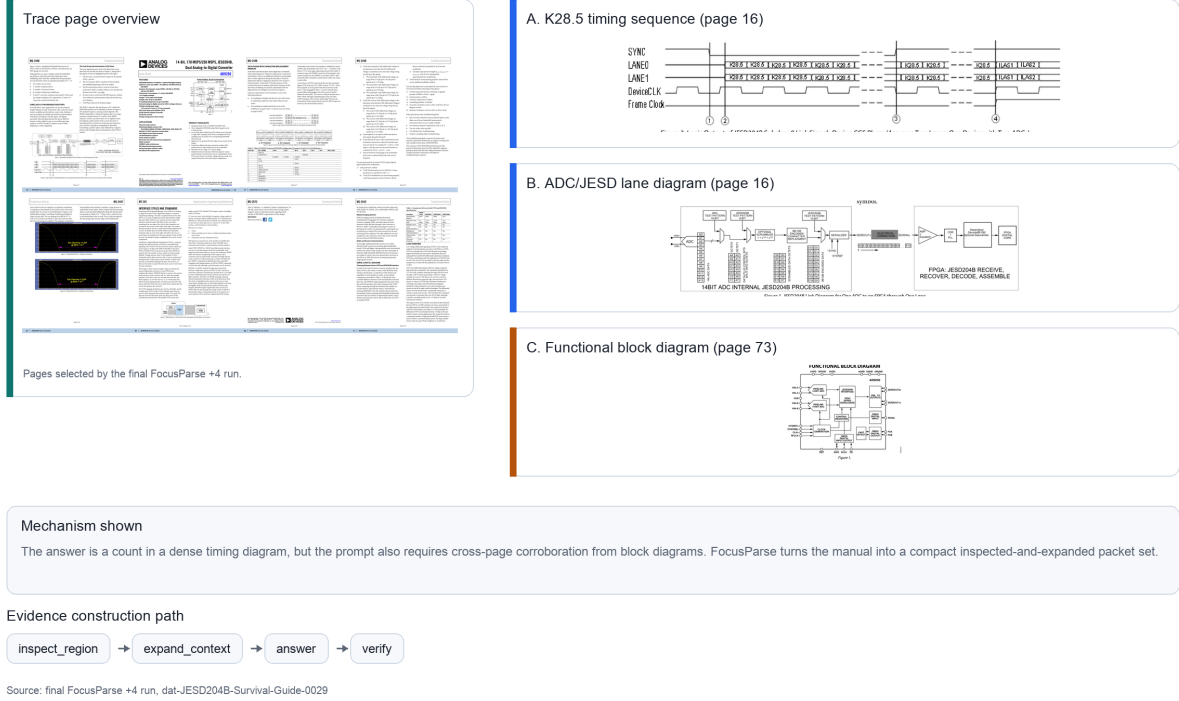


Figure 2: Datasheet evidence compaction. Several baselines can state or nearly state the integer answer, but they fail to preserve complete page/box grounding. FocusParse +4 answers 6 with the timing diagram and cross-page corroborating block diagrams retained as citable evidence packets.

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A Artifact Provenance

The main result table, composed qualitative panels, and same-revision qualitative baseline predictions are sourced from the May 24 result package recorded in the paper-draft runbook. A slim review archive is available at [results/paper/submission-review-package/](#); the exact archive path is recorded in the package manifest. All claims in the abstract use that final matched package.